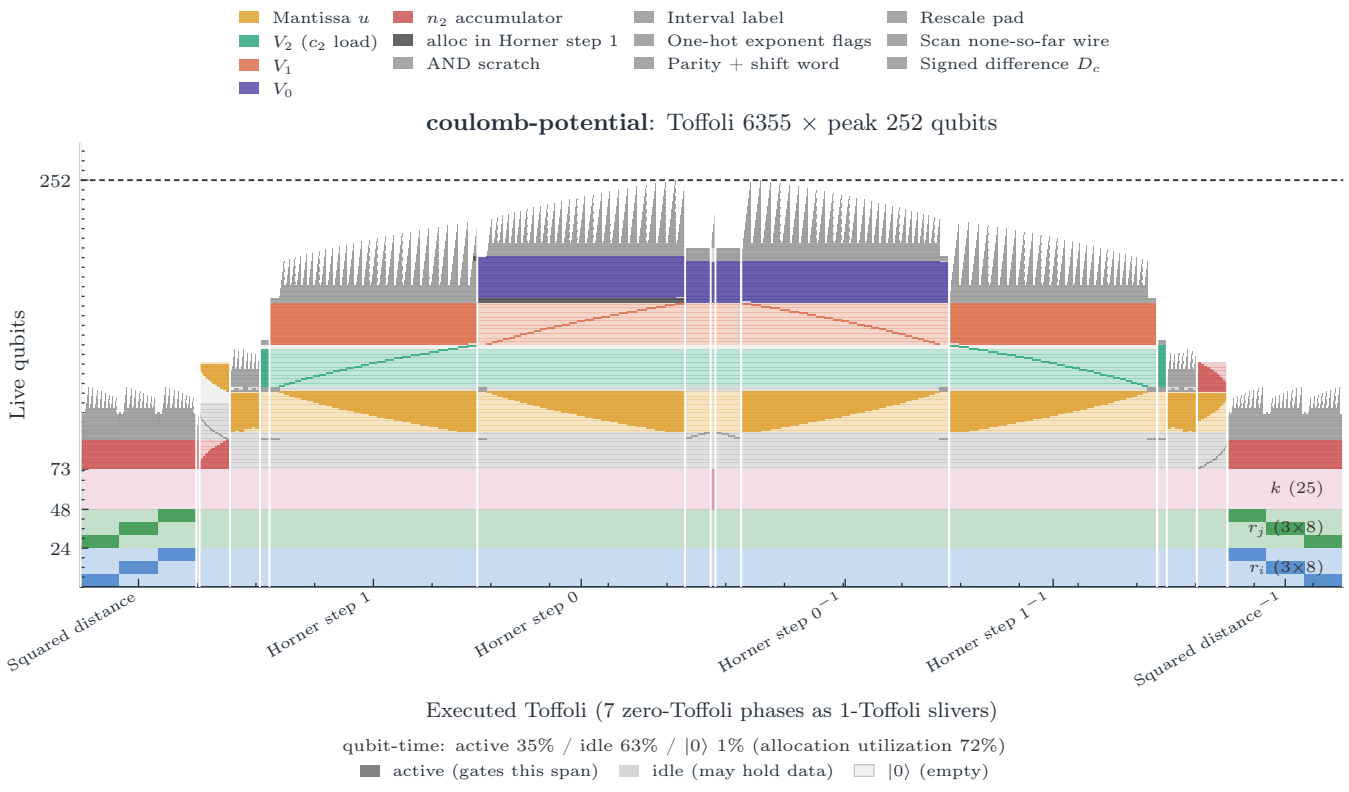

Mantissa u
V_2 (c_2 load)
V_1
V_0
n_2 accumulator
alloc in Horner step 1
AND scratch
Interval label
One-hot exponent flags
Parity + shift word
Rescale pad
Scan none-so-far wire
Signed difference D_c
coulomb-potential: Toffoli 6355 × peak 252 qubits
Live qubits
252
73
48
24
k (25)
r_j (3×8)
r_i (3×8)
Squared distance
Horner step 1
Horner step 0
Horner step 0−1
Horner step 1−1
Squared distance−1
Executed Toffoli (7 zero-Toffoli phases as 1-Toffoli slivers)
qubit-time: active 35% / idle 63% / |0⟩ 1% (allocation utilization 72%)
active (gates this span)
idle (may hold data)
|0⟩ (empty)